

3.9 Exercises

Exercise 3.1: Removing the outer min in a min-max problem

Show that $V_i^0 : \mathcal{X}_i \rightarrow \mathbb{R}$ and $\kappa_i : \mathcal{X}_i \rightarrow \mathbb{U}$ defined by

$$\begin{aligned} V_i^0(x) &= \min_{u \in \mathbb{U}} \max_{w \in \mathbb{W}} \{ \ell(x, u, w) + V_{i-1}^0(f(x, u, w)) \mid f(x, u, \mathbb{W}) \subset \mathcal{X}_{i-1} \} \\ \kappa_i(x) &= \arg \min_{u \in \mathbb{U}} \max_{w \in \mathbb{W}} \{ \ell(x, u, w) + V_{i-1}^0(f(x, u, w)) \mid f(x, u, \mathbb{W}) \subset \mathcal{X}_{i-1} \} \\ \mathcal{X}_i &= \{x \in \mathbb{X} \mid \exists u \in \mathbb{U} \text{ such that } f(x, u, \mathbb{W}) \subset \mathcal{X}_{i-1}\} \end{aligned}$$

satisfy

$$V_i^0(x) = \max_{w \in \mathbb{W}} \{ \ell(x, \kappa_i(x), w) + V_{i-1}^0(f(x, \kappa_i(x), w)) \}$$

Exercise 3.2: Maximizing a difference

Prove the claim used in the proof of Theorem 3.9 that

$$\max_w \{a(w)\} - \max_w \{b(w)\} \leq \max_w \{a(w) - b(w)\}$$

Also show the following minimization version

$$\min_w \{a(w)\} - \min_w \{b(w)\} \geq \min_w \{a(w) - b(w)\}$$

Exercise 3.3: Equivalent constraints

Assuming that S is a polytope and, therefore, defined by linear inequalities, show that the constraint $x \in \{z\} \oplus S$ (on z for given x) may be expressed as $Bz \leq b + Bx$, i.e., z must lie in a polytope. If S is symmetric ($x \in S$ implies $-x \in S$), show that $x \in \{z\} \oplus S$ is equivalent to $z \in \{x\} \oplus S$.

Exercise 3.4: Hausdorff distance between translated sets

Prove that the Hausdorff distance between two sets $\{x\} \oplus S$ and $\{y\} \oplus S$, where S is a compact subset of $\mathbb{R}^n$ and x and y are points in $\mathbb{R}^n$, is $|x - y|$.

Exercise 3.5: Exponential convergence of $X(i)$

Complement the proof of Proposition 3.12 by proving the sequence of sets $(X(i))_{0:\infty}$, $X(i) := \{\tilde{x}(i)\} \oplus S_K(\infty)$, converges exponentially fast to the set $S_K(\infty)$ as $i \rightarrow \infty$ if $\tilde{x}(i)$ converges exponentially fast to 0 as $i \rightarrow \infty$.

Exercise 3.6: Simulating a robust MPC controller

This exercise explores robust MPC for linear systems with an additive bounded disturbance

$$x^+ = Ax + Bu + w$$

The first task, using the tube-based controller described in Section 3.5.3 is to determine state and control constraint sets $\mathbb{Z}$ and $\mathbb{V}$ such that if the nominal system $z^+ = Az + Bv$ satisfies $z \in \mathbb{Z}$ and $v \in \mathbb{V}$, then the actual system $x^+ = Ax + Bu + w$ with $u = v + K(x - z)$ where K is such that $A + BK$ is strictly stable, satisfies the constraints $x \in \mathbb{X}$ and $u \in \mathbb{U}$.

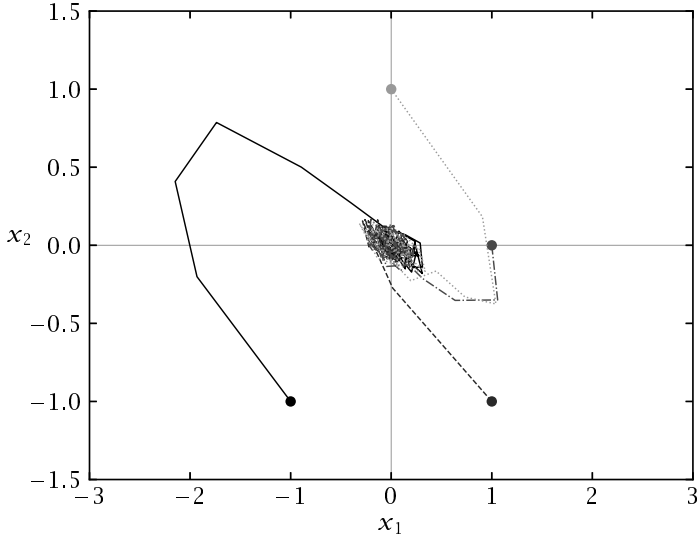


Figure 3.10: Closed-loop robust MPC state evolution with uniformly distributed $|w| \leq 0.1$ from four different x_0 .

- (a) To get started, consider the scalar system

$$x^+ = x + u + w$$

with constraint sets $\mathbb{X} = \{x \mid x \leq 2\}$, $\mathbb{U} = \{u \mid |u| \leq 1\}$, and $\mathbb{W} = \{w \mid |w| \leq 0.1\}$. Choose $K = -(1/2)$ so that $A_K = 1/2$. Determine $\mathbb{Z}$ and $\mathbb{V}$ so that if the nominal system $z^+ = z + v$ satisfies $z \in \mathbb{Z}$ and $v \in \mathbb{V}$, the uncertain system $x^+ = Ax + Bu + w$, $u = v + K(x - z)$ satisfies $x \in \mathbb{X}$, $u \in \mathbb{U}$.

- (b) Repeat part (a) for the following uncertain system

$$x^+ = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u + w$$

with the constraint sets $\mathbb{X} = \{x \in \mathbb{R}^2 \mid x_1 \leq 2\}$, $\mathbb{U} = \{u \in \mathbb{R} \mid |u| \leq 1\}$ and $\mathbb{W} = [-0.1, 0.1]$. Choose $K = \begin{bmatrix} -0.4 & -1.2 \end{bmatrix}$.

- (c) Determine a model predictive controller for the nominal system and constraint sets $\mathbb{Z}$ and $\mathbb{V}$ used in (b).
- (d) Implement robust MPC for the uncertain system and simulate the closed-loop system for a few initial states and a few disturbance sequences for each initial state. The phase plot for initial states $[-1, -1]$, $[1, 1]$, $[1, 0]$, and $[0, 1]$ should resemble Figure 3.10.

Bibliography

- D. A. Allan, C. N. Bates, M. J. Risbeck, and J. B. Rawlings. On the inherent robustness of optimal and suboptimal nonlinear MPC. *Sys. Cont. Let.*, 106: 68–78, August 2017.
- D. Angeli, A. Casavola, G. Franze, and E. Mosca. An ellipsoidal off-line MPC scheme for uncertain polytopic discrete-time systems. *Automatica*, 44: 3113–3119, 2008.
- J. P. Aubin. *Viability Theory*. Systems & Control: Foundations & Applications. Birkhauser, Boston, Basel, Berlin, 1991.
- D. P. Bertsekas and I. B. Rhodes. Recursive state estimation for a set-membership description of uncertainty. *IEEE Trans. Auto. Cont.*, 16:117–128, 1971a.
- D. P. Bertsekas and I. B. Rhodes. On the minimax reachability of target sets and target tubes. *Automatica*, 7(2):233–247, 1971b.
- F. Blanchini. Set invariance in control. *Automatica*, 35:1747–1767, 1999.
- F. Blanchini and S. Miani. *Set-Theoretic methods in Control*. Systems & Control: Foundations and Applications. Birkhäuser, 2008.
- G. C. Calafiore and M. C. Campi. The scenario approach to robust control design. *IEEE Trans. Auto. Cont.*, 51(5):742–753, May 2006.
- G. C. Calafiore and L. Fagiano. Robust model predictive control via scenario optimization. *IEEE Transactions on Automatic Control*, 56(1):219–224, January 2013.
- D. Chatterjee and J. Lygeros. On stability and performance of stochastic predictive control techniques. *IEEE Trans. Auto. Cont.*, 60(2):509–514, 2015.
- D. Chatterjee, P. Hokayem, and J. Lygeros. Stochastic receding horizon control with bounded control inputs: a vector space approach. *IEEE Trans. Auto. Cont.*, 56(11):2704–2710, November 2011.
- H. Chen and F. Allgöwer. A quasi-infinite horizon nonlinear model predictive control scheme with guaranteed stability. *Automatica*, 34(10):1205–1217, 1998.

- L. Chisci, J. A. Rossiter, and G. Zappa. Systems with persistent disturbances: predictive control with restricted constraints. *Automatica*, 37(7):1019–1028, 2001.
- G. De Nicolao, L. Magni, and R. Scattolini. Robust predictive control of systems with uncertain impulse response. *Automatica*, 32(10):1475–1479, 1996.
- M. Diehl, H. G. Bock, and E. Kostina. An approximation technique for robust nonlinear optimization. *Math. Prog.*, 107:213–230, 2006.
- P. Falugi and D. Q. Mayne. Tube-based model predictive control for nonlinear systems with unstructured uncertainty. In *Proceedings of 50th IEEE Conference on Decision and Control*, pages 2656–2661, Orlando, Florida, USA, December 2011.
- P. J. Goulart, E. C. Kerrigan, and J. M. Maciejowski. Optimization over state feedback policies for robust control with constraints. *Automatica*, 42:523–533, 2006.
- P. J. Goulart, E. C. Kerrigan, and D. Ralph. Efficient robust optimization for robust control with constraints. *Math. Prog.*, 114(1):115–147, July 2008.
- G. Grimm, M. J. Messina, S. E. Tuna, and A. R. Teel. Nominally robust model predictive control with state constraints. *IEEE Trans. Auto. Cont.*, 52(10):1856–1870, October 2007.
- G. A. Hicks and W. H. Ray. Approximation methods for optimal control synthesis. *Can. J. Chem. Eng.*, 49:522–528, August 1971.
- S. Kameswaran and L. T. Biegler. Simultaneous dynamic optimization strategies: Recent advances and challenges. *Comput. Chem. Eng.*, 30:1560–1575, September 2006.
- C. M. Kellet and A. R. Teel. Discrete-time asymptotic controllability implies smooth control-Lyapunov function. *Sys. Cont. Let.*, 52:349–359, 2004.
- A. B. Khurzhanski and I. Valyi. *Ellipsoidal-valued dynamics for estimation and control*. Systems & Control: Foundations & Applications. Birkhauser, Boston, Basel, Berlin, 1997.
- I. Kolmanovsky and E. G. Gilbert. Maximal output admissible sets for discrete-time systems with disturbance inputs. In *Proceedings of the American Control Conference*, Seattle, June 1995.
- I. Kolmanovsky and E. G. Gilbert. Theory and computation of disturbance invariant sets for discrete-time linear systems. *Math. Probl. Eng.*, 4(4):317–367, 1998.

- M. V. Kothare, V. Balakrishnan, and M. Morari. Robust constrained model predictive control using linear matrix inequalities. *Automatica*, 32(10):1361–1379, 1996.
- K. I. Kouramas, S. V. Raković, E. C. Kerrigan, J. C. Allwright, and D. Q. Mayne. On the minimal robust positively invariant set for linear difference inclusions. In *Proceedings of the 44th IEEE Conference on Decision and Control and European Control Conference ECC 2005*, pages 2296–2301, Sevilla, Spain, December 2005.
- B. Kouvaritakis, M. Cannon, S. V. Raković, and Q. Cheng. Explicit use of probabilistic distributions in linear predictive control. *Automatica*, 46:1719–1724, 2010.
- W. Langson, I. Chrysoschoos, S. V. Raković, and D. Q. Mayne. Robust model predictive control using tubes. *Automatica*, 40:125–133, January 2004.
- M. Lazar, D. M. de la Peña, W. P. M. H. Heemels, and T. Alamo. On input-to-state stability of min-max nonlinear model predictive control. *Sys. Cont. Let.*, 57: 39–48, 2008.
- J. H. Lee and Z. Yu. Worst-case formulations of model predictive control for systems with bounded parameters. *Automatica*, 33(5):763–781, 1997.
- D. Limon, T. Alamo, and E. F. Camacho. Stability analysis of systems with bounded additive uncertainties based on invariant sets: stability and feasibility of MPC. In *Proceedings of the American Control Conference*, pages 364–369, Anchorage, Alaska, May 2002.
- D. Limon, T. Alamo, D. M. Raimondo, D. M. de la Peña, J. M. Bravo, and E. F. Camacho. Input-to-state stability: an unifying framework for robust model predictive control. In L. Magni, D. M. Raimondo, and F. Allgöwer, editors, *International Workshop on Assessment and Future Directions of Nonlinear Model Predictive Control*, Pavia, Italy, September 2008.
- M. Lorenzen, F. Dabbene, R. Tempo, and F. Allgöwer. Constraint-tightening and stability in stochastic model predictive control. *arXiv:1511.03488v2 [cs.SY]*, 2015.
- C. Løvaas, M. M. Serón, and G. C. Goodwin. Robust output feedback model predictive control for systems with unstructured uncertainty. *Automatica*, 44(8):1933–1943, August 2008.
- L. Magni and R. Sepulchre. Stability margins of nonlinear receding-horizon control via inverse optimality. *Sys. Cont. Let.*, 32:241–245, 1997.

- L. Magni, G. De Nicolao, R. Scattolini, and F. Allgöwer. Robust model predictive control for nonlinear discrete-time systems. *Int. J. Robust and Nonlinear Control*, 13:229–246, 2003.
- D. Q. Mayne. Optimization in model based control. In *Proceedings of the IFAC Symposium Dynamics and Control of Chemical Reactors, Distillation Columns and Batch Processes*, pages 229–242, Helsingor, Denmark, June 1995.
- D. Q. Mayne. Nonlinear model predictive control: An assessment. In J. C. Kantor, C. E. García, and B. Carnahan, editors, *Proceedings of Chemical Process Control – V*, pages 217–231. CACHE, AIChE, 1997.
- D. Q. Mayne. Robust and stochastic model predictive control: Are we going in the right direction? *Annual Rev. Control*, 2016.
- D. Q. Mayne and W. Langson. Robustifying model predictive control of constrained linear systems. *Electron. Lett.*, 37(23):1422–1423, 2001.
- D. Q. Mayne, J. B. Rawlings, C. V. Rao, and P. O. M. Scokaert. Constrained model predictive control: Stability and optimality. *Automatica*, 36(6):789–814, 2000.
- D. Q. Mayne, M. M. Serón, and S. V. Raković. Robust model predictive control of constrained linear systems with bounded disturbances. *Automatica*, 41(2):219–224, February 2005.
- D. Q. Mayne, E. C. Kerrigan, E. J. van Wyk, and P. Falugi. Tube based robust nonlinear model predictive control. *Int. J. Robust and Nonlinear Control*, 21(11):1341–1353, 2011.
- Z. Nagy and R. Braatz. Open-loop and closed-loop robust optimal control of batch processes using distributional and worst-case analysis. *J. Proc. Cont.*, pages 411–422, 2004.
- G. Pannocchia, J. B. Rawlings, and S. J. Wright. Conditions under which suboptimal nonlinear MPC is inherently robust. *Sys. Cont. Let.*, 60:747–755, 2011.
- S. V. Raković. Invention of prediction structures and categorization of robust MPC syntheses. In *Proceedings of 4th IFAC Nonlinear Model Predictive Control Conference*, pages 245–273, Noordwijkerhout, NL, August 2012.
- S. V. Raković, E. C. Kerrigan, K. I. Kouramas, and D. Q. Mayne. Approximation of the minimal robustly positively invariant set for discrete-time LTI systems with persistent state disturbances. In *Proceedings 42nd IEEE Conference on Decision and Control*, volume 4, pages 3917–3918, Maui, Hawaii, USA, December 2003.

- S. V. Raković, E. C. Kerrigan, K. I. Kouramas, and D. Q. Mayne. Invariant approximations of the minimal robustly positively invariant sets. *IEEE Trans. Auto. Cont.*, 50(3):406–410, 2005a.
- S. V. Raković, D. Q. Mayne, E. C. Kerrigan, and K. I. Kouramas. Optimized robust control invariant sets for constrained linear discrete-time systems. In *Proceedings of 16th IFAC World Congress on Automatic Control*, Prague, Czechoslovakia, 2005b.
- S. V. Raković, B. Kouvaritakis, M. Cannon, C. Panos, and R. Findeisen. Parameterized tube model predictive control. *IEEE Trans. Auto. Cont.*, 57(11): 2746–2761, 2012.
- J. A. Rossiter, B. Kouvaritakis, and M. J. Rice. A numerically robust state-space approach to stable-predictive control strategies. *Automatica*, 34(1):65–73, 1998.
- L. O. Santos and L. T. Biegler. A tool to analyze robust stability for model predictive control. *J. Proc. Cont.*, 9:233–245, 1999.
- P. O. M. Scokaert and D. Q. Mayne. Min-max feedback model predictive control for constrained linear systems. *IEEE Trans. Auto. Cont.*, 43(8):1136–1142, August 1998.
- P. O. M. Scokaert, J. B. Rawlings, and E. S. Meadows. Discrete-time stability with perturbations: Application to model predictive control. *Automatica*, 33(3): 463–470, 1997.
- A. R. Teel. Discrete time receding horizon control: is the stability robust. In Marcia S. de Queiroz, Michael Malisoff, and Peter Wolenski, editors, *Optimal control, stabilization and nonsmooth analysis*, volume 301 of *Lecture notes in control and information sciences*, pages 3–28. Springer, 2004.
- R. Tempo, G. C. Calafiore, and F. Dabbene. *Randomized algorithms for analysis and control of uncertain systems: With applications*. Springer, second edition, 2013.
- S. Yu, M. Reble, H. Chen, and F. Allgöwer. Inherent robustness properties of quasi-infinite horizon MPC. In *Proceedings of the 18th IFAC World Congress*, Milano, Italy, August, September 2011.